

Computer Science & IT

Database Management System



Relational Model & Normal Forms

Lecture No. 14



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Recap of Previous Lecture

✓ Topic

Decomposition of relation



Topics to be Covered



✓ Topic

Decomposition of relation

✓ Topic

Multivalued dependency

✓ Topic

Properties of multivalued dependency

✓ Topic

Fourth normal form (4NF)

#e.g. Given $R(ABCDE)$ and $F:\{A \rightarrow BC, CD \rightarrow E, B \rightarrow D, E \rightarrow A\}$

Find the normal form of the relation, and if relation is not already in BCNF then decompose the relation up to BCNF.

C.K = A, E, CD, BC

✓ $\underset{\text{S.K}}{A} \rightarrow BC \quad \{BCNF\}$

✓ $\underset{\text{S.K}}{CD} \rightarrow E \quad \{BCNF\}$

✓ $\underset{\text{S.K}}{E} \rightarrow A \quad \{BCNF\}$

✓ $\underset{\text{P.S. of one C.K}}{B} \xrightarrow{\text{Case 4}} \underset{\text{Proper subset of another C.K}}{D} \quad \{3NF\}$

$R(ABCDE)$ is not in BCNF becoz of $B \rightarrow D$
∴ Decompose w.r.t $B \rightarrow D$

$$(BD)^+ = \{BD\}$$

$R_1(BD)$

$$F_1 = \{ B \rightarrow D \}$$

$PK = B$

BCNF

BCNF

$R_2(ABCE)$

$$F_2 = \begin{cases} A \rightarrow BCE \\ E \rightarrow ABC \\ BC \rightarrow AE \end{cases}$$

$PK = A, (BC), E$

for lossless

BCNF

To preserve $CD \rightarrow E$
We can create a
new subrelation

$R_3(CDE)$

$$F_3 = \{ CD \rightarrow E \}$$

$PK = CD$

$$(CD)^+ \text{ w.r.t. } F_1 \cup F_2 = \{CD\}$$

∴ $CD \rightarrow E$ will be
lost.

BCNF
+
Lossless Join
+
Dep. preserving



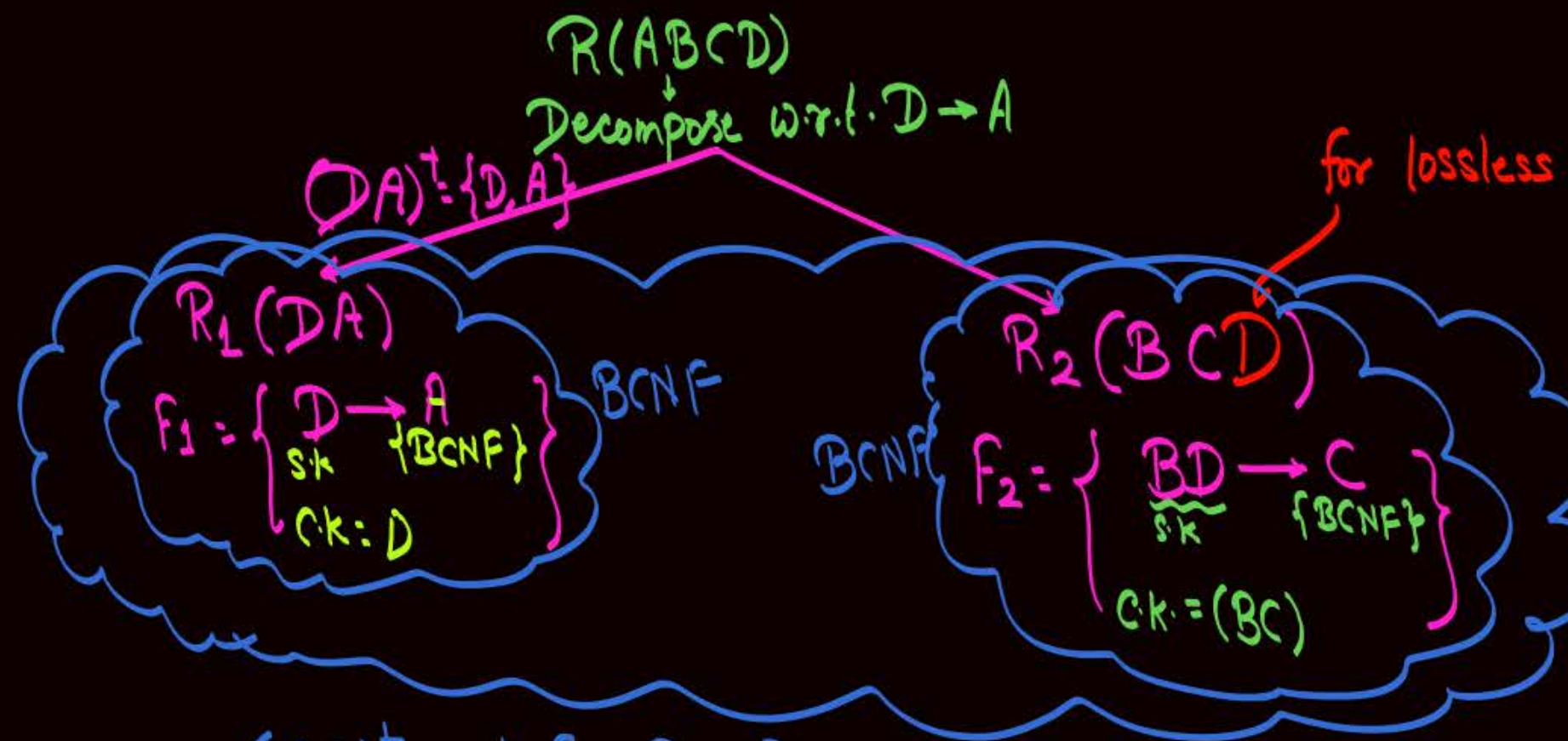
Given $R(ABCD)$ and $F=\{AB \rightarrow CD, D \rightarrow A\}$

Find the normal form of the relation, and if relation is not already in BCNF then decompose the relation up to BCNF.

C.K. = AB, DB

$\underbrace{AB}_{S.K.} \rightarrow CD \{BCNF\}$

$\underbrace{D}_{P.S.C.K.} \rightarrow \underbrace{A}_{P.S.C.K.} \{3NF\}$



Overall normal form of DB = BCNF
+ Lossless join

But ⁺ not dependency preserving

$(AB)^+$ w.r.t. $F_1 \cup F_2 = \{A, B\}$

$\therefore AB \rightarrow CD$ is not implied by $F_1 \cup F_2$

└ If we create a subrelation $R'(ABCD)$ to preserve $AB \rightarrow CD$ then $D \rightarrow A$ will also exist in R' and $D \rightarrow A$ will be in 3NF

└ i.e., we will not be able to preserve $AB \rightarrow CD$ while ensuring that the Normal form improves to BCNF

Note:-

- ① Upto 3NF we can always ensure Lossless Join and dependency Preserving decomposition.
- ② While decomposing a relation into BCNF we can always ensure that the decomposition is lossless Join decomposition, but when decomposing a relation into BCNF some times (not always) it may not be possible to preserve some of the functional dependencies of the original relation.
- ③ Most adequate normal form of the database is 3NF, because upto 3NF we can ensure both lossless join decomposition as well as dependency Preserving decomposition.

④ In a relation if all attributes of the relation are prime attributes, then that relation is at least in 3NF.

⑤ If all candidate keys of the relation are simple candidate keys, then partial dependency can not exist in that relation, and hence the relation will be at least in 2NF.

⑥ In a relation, if all attributes of the relation are prime attributes, and all candidate keys of the relation are simple candidate keys, then that relation is in BCNF.

⑦ A binary relation {i.e. A relation with exactly two attributes} is always in BCNF.

Let $R(AB)$ is a relation,
then Possible FD sets of R are

$\left\{ \begin{array}{l} A \rightarrow B \\ B \rightarrow A \end{array} \right\}$ {BCNF}
C.K = $(A) \& (B)$

$\{ A \rightarrow B \}$ {BCNF}
C.K = A

$\{ B \rightarrow A \}$ {BCNF}
C.K = B

$\{ AB \rightarrow AB \}$
i.e. no non-trivial FD
 $\therefore$ BCNF

⑧ If a relation is in BCNF, then there will be 0% redundancy in that relation because of functional dependency, but such relation may still suffer from redundancy present because of Multi-valued dependency

Sid	Cid	Proj-No.
S ₁	C ₁	P ₁
S ₁	C ₁	P ₂
S ₁	C ₂	P ₁
S ₁	C ₂	P ₂
S ₂	C ₂	P ₂

Redundancy
Exist

it exist because of
Multi-valued dependency

$Sid \not\rightarrow Cid$
 $Sid \not\rightarrow Proj-No.$
 $Cid \not\rightarrow Sid$
 $Cid \not\rightarrow Proj-No.$
 $Proj-No. \not\rightarrow Sid$
 $Proj-No. \not\rightarrow Cid$
 $Sid, Cid \not\rightarrow Proj-No.$
 $Sid, Proj-No. \not\rightarrow Cid$
 $Cid, Proj-No. \not\rightarrow Sid$

i.e.
 No non-trivial
 functional
 dependency
 exists in
 relation.
 ∴ Relation is
 at least in
 BCNF



Topic : Multivalued dependency

"X multivalued determines Y" is denoted by $X \twoheadrightarrow Y$

- ⊗ If there exist two or more independent attributes which are dependent on some other set of attributes then multi-valued dependency will exist in the relation

Formal definition of MVD:

Let R be the relation, and X & Y are some set of attributes from relation R ,

{ let us define
 $Z = \text{Attributes}_{\text{of } R} - (X \cup Y)$

If there exists at least 4 tuples $t_1, t_2, t_3, t_4 \in R$ such that,

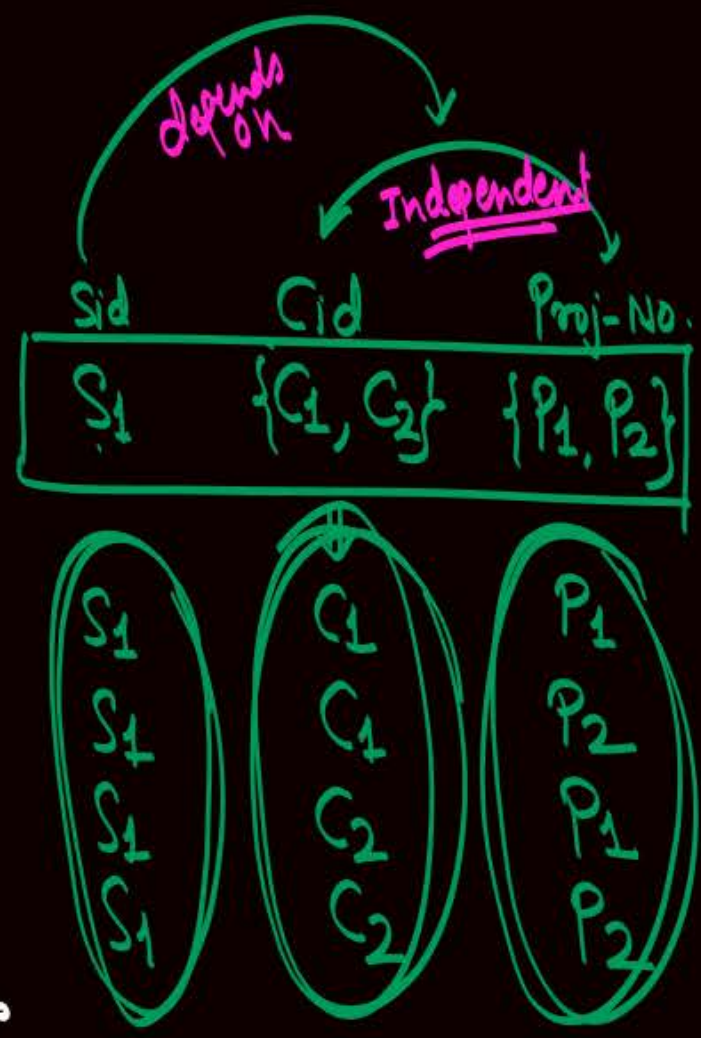
$$\left\{ \begin{array}{l} t_1.X = t_2.X = t_3.X = t_4.X \\ \text{and} \\ t_1.Y = t_2.Y \text{ and } t_3.Y = t_4.Y \\ \text{and} \\ t_1.Z = t_3.Z \text{ and } t_2.Z = t_4.Z \end{array} \right.$$

i.e., Z is a set of remaining attributes of R

then Multi-valued dependency, $X \twoheadrightarrow Y$ exists in R .

then $Sid \twoheadrightarrow Proj-No.$ is also valid.

	X Sid	Y Cid	Z Proj-No.
t_1	S_1	C_1	P_1
t_3	S_1	C_1	P_2
t_2	S_1	C_2	P_1
t_4	S_1	C_2	P_2
	S_2	C_2	P_2



We can observe $\{ X = Sid, Y = Cid, Z = Proj-No. \}$

$$\left. \begin{aligned}
 &t_1.X = t_2.X = t_3.X = t_4.X \\
 &\text{and} \\
 &t_1.Y = t_2.Y \text{ and } t_3.Y = t_4.Y \\
 &\text{and} \\
 &t_1.Z = t_3.Z \text{ and } t_2.Z = t_4.Z
 \end{aligned} \right\}$$

$\Rightarrow \therefore X \twoheadrightarrow Y$
 ie, $Sid \twoheadrightarrow Cid$ exists in Relation

* Another definition w.r.t. MVD :-

Whenever we swap the values of attributes of set Y in two tuples (let t_1, t_2) which agree on the values of the attributes of set X {i.e. $t_1.X = t_2.X$ }, and if the resulting tuples were already present in the relation,

then $X \twoheadrightarrow Y$ in that relation
exist

	X Sid	Y Cid	Z Proj-No.
t ₁	S ₁	C₁	P ₁
t ₂	S ₁	C ₁	P ₂
t ₃	S ₁	C₂	P ₁
t ₄	S ₁	C ₂	P ₂
	S ₂	C ₂	P ₂

Resulting tuples
were already
present in
the relation

Does $Cid \twoheadrightarrow Sid$ exist in relation
 let $t_1 = (S_1, C_2, P_1)$
 & $t_2 = (S_2, C_2, P_2)$
 agree of the value of Cid
 ∴ let us swap the values of Sid
 ∴ Resulting tuples are

Not already
a member
 ∴ $Cid \twoheadrightarrow Sid$
 does not
hold true
in the
relation



2 mins Summary



Topic

Decomposition of relation

Topic

Multivalued dependency

Topic

Properties of multivalued dependency

Topic

Fourth normal form (4NF)

THANK - YOU